

CONSTRUCTION OS

PRD - BIM, 3D & Digital Twin

Full Product Requirements Document for Product, UX/UI, Frontend, Backend, BIM Engineering, Data, DevOps, QA, Security and Platform Operations Teams

Product purpose: Build the premium visual coordination and digital twin layer that connects every model object to governed ConstructionOS data while preserving each operational module as the source of truth.

Document control	Value
Document	PRD_BIM_3D_Digital_Twin
Version	1.0
Status	Implementation-ready functional and technical specification
Audience	Full-stack, WebGL/WebGPU, BIM, data, platform, security, QA, DevOps, SRE and product teams
Primary routes	/bim, /projects/{projectId}/digital-twin, /bim/models, /bim/federations, /bim/admin
Core architecture	Controlled model registry + asynchronous conversion pipeline + stable object identity + federated scene graph + browser viewer + governed cross-module links
Access model	Deny-by-default, tenant-isolated, company/project scoped, model/layer/object restricted and link-target permission checked
Source-of-truth policy	BIM owns model records, versions, federations, object identity and visual relationships; connected modules own their operational records; Documents owns controlled source files
Integration policy	Open-format first. IFC, glTF/GLB and BCF are core targets; Revit, Navisworks, Unreal Engine, Cesium and future ROZARIS integrations use adapters and versioned APIs
Platform Configuration	Mandatory at project, model, federation, discipline, layer, object class, viewer tool, link type, processing profile and export level
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Table of Contents

1. Executive Summary
2. Product Vision, Outcomes and Failure Risks
3. Scope, Non-Goals and Assumptions
4. Terminology and Core Concepts
5. Permanent Source-of-Truth Boundaries
6. Personas, Roles and Access Model
7. Information Architecture
8. Digital Twin Spatial Hierarchy
9. Model Registry and Model Lifecycle
10. Supported Formats and Integration Posture
11. Model Ingestion and Validation
12. Coordinate Systems, Georeferencing and Units
13. Model Versioning and Revision Control
14. Model Federation and Discipline Coordination
15. Conversion, Optimization and Publishing Pipeline
16. Stable Object Identity and Object Matching
17. Classification, Property Sets and Data Normalization
18. Browser Viewer Architecture
19. Viewer Navigation and Camera Controls
20. Object Selection, Search, Filters and Isolation
21. Layers, Disciplines, Worksets and Visibility
22. Sections, Clipping, Explode and Measurements
23. Viewpoints, Markups and Presentation States
24. Object Data Panel and Relationship Graph
25. BCF, Coordination Issues and Clash Records
26. Tasks, Comments, Meetings and Decisions Integration
27. Documents, Drawings and Submittals Integration
28. Work Progress Integration and Visual Status
29. 4D Construction Sequence and Baseline Comparison
30. QA/QC, Inspections, NCRs and Handover Integration
31. HSE, Permits, Risks and Emergency Context
32. Assets, Equipment and Operational Digital Twin
33. Units, Sales and Commercial Visualization
34. Quantities, BOQ, Cost and 5D Readiness
35. Reality Capture, Point Clouds and Change Comparison
36. Geospatial, Cesium and Site Context
37. Unreal Engine, ArchViz and ROZARIS Readiness
38. Digital Twin State, Telemetry and IoT Readiness
39. Change Detection and Model Comparison
40. Permissions, Confidential Layers and External Sharing
41. Notifications, Events and Scheduled Processing
42. Platform Configuration and Administration
43. Data Model
44. Backend Service Architecture
45. Processing Workers and Storage Architecture
46. Domain Events and Integration Contracts
47. REST and Streaming API Contract
48. Frontend Architecture and State Management
49. Search Indexing and Query Architecture
50. Mobile, Tablet and Offline Behavior
51. Security, Privacy and Intellectual Property
52. Audit, Observability and Operations
53. Performance, Scalability and Availability
54. Validation, Error Handling and Concurrency
55. Retention, Archiving, Legal Hold and Deletion
56. Testing Strategy and Critical Scenarios
57. Acceptance Criteria
58. Delivery Phases
59. Migration and Adoption
60. Page and Flow Registry
61. Risks and Open Decisions
62. Definition of Done

1. Executive Summary

The BIM, 3D & Digital Twin Module is the premium visual coordination layer of ConstructionOS. It transforms project models into a governed, browser-accessible spatial interface where every persistent model object can be connected to tasks, work progress, drawings, documents, RFIs, inspections, defects, materials, contractors, costs, assets, units, risks, HSE records, comments, meetings and decisions.

The module is not a standalone model viewer and must not become a second database for operational information. It owns model registration, conversion, versions, federations, stable object identity, normalized properties, viewpoints, markups and visual relationships. Connected modules retain ownership of their own records and are opened through permission-safe projections and deep links.

Permanent rule: BIM owns the visual and spatial truth of model versions and objects. Operational modules own business facts. The Digital Twin connects them through stable, auditable references.

1.1 Required outcomes

- Open large federated project models in a web browser without requiring authoring software.
- Maintain stable object references across model revisions wherever matching confidence is sufficient.
- Navigate from company and project to building, block, level, zone, unit, room, system and individual element.
- Expose authorized operational context on model objects without copying authoritative records.
- Visualize planned, actual, accepted, certified and blocked work states distinctly.
- Support coordination through BCF-compatible issues, viewpoints, markups and clash references.
- Provide performant desktop, tablet and mobile viewing with model LOD, streaming and caching.
- Remain integration-ready for Revit, Navisworks, Unreal Engine, Cesium and future ROZARIS technology.

1.2 Success indicators

Outcome	Target behavior
Model access	Authorized users open published project models in-browser with no desktop BIM dependency.
Traceability	Any visual state can be traced to a model version, object reference and source operational record.
Revision continuity	Stable object links survive model revision when identity/matching rules confirm continuity.
Performance	Initial spatial context loads quickly while detailed geometry streams progressively.
Security	Restricted models, layers, objects and linked records are not inferable through search, exports or previews.
Reliability	Failed conversion, federation or tile generation is isolated, observable and safely retryable.

2. Product Vision, Outcomes and Failure Risks

2.1 Product vision

ConstructionOS should let a user select any meaningful physical object in a project and immediately understand its design identity, location, current construction state and connected business context. The Digital Twin becomes the shared spatial language between design, construction, commercial, quality, safety, facilities and sales teams.

2.2 Business outcomes

- Faster coordination between design, site, commercial and management teams.
- Reduced time spent locating drawings, issues, inspections and work records.
- Clearer progress reviews through model-based visual states.

- Lower risk of working from obsolete model revisions.
- Consistent object-level traceability across the project lifecycle.
- Better handover readiness and future asset operations.
- Premium client, developer and executive experiences without compromising source-of-truth discipline.

2.3 Failure risks the architecture must prevent

Risk	Required prevention
Viewer becomes a file preview only	Object graph, operational links, lifecycle states and governed workflows are first-class.
Object links break after every revision	Stable IDs, deterministic matching, confidence scoring and manual reconciliation.
Model data replaces operational data	Read-only projections and deep links; source modules retain writes and decisions.
Large model crashes browsers	Segmentation, LOD, mesh compression, progressive streaming, memory budgets and worker-based parsing.
Restricted information leaks in 3D	Model/layer/object permission checks plus target-record permission checks.
Coordinate misalignment creates false decisions	Explicit coordinate reference, unit, transform and validation controls.
Derived quantities presented as certified truth	Clear provenance, tolerance, version and approval status.
Vendor lock-in	Open-format first, adapter-based proprietary integrations and exportable metadata.

3. Scope, Non-Goals and Assumptions

3.1 In scope

- Model registry, upload/reference, validation, conversion, optimization, publication and archival.
- IFC, glTF/GLB and BCF core workflows; proprietary authoring-tool adapters through controlled integration boundaries.
- Federated models, discipline coordination, object identity, hierarchy, search and normalized properties.
- Web viewer controls, sections, measurements, viewpoints, markups, comparison and status coloring.
- Links to Projects, Units, Tasks, Documents, Work Progress, QA/QC, HSE, Assets, Finance/BOQ references, Contracts and Meetings.
- 4D sequence visualization and 5D readiness without duplicating schedule or financial source records.
- Geospatial context, reality capture readiness and digital-twin telemetry readiness.
- Administration, APIs, events, audit, observability, security, responsive behavior and testing.

3.2 Non-goals

- Replacing Revit, Archicad, Tekla, Civil 3D, Navisworks or other authoring/coordination applications.
- Editing native BIM geometry or authoring design models in-browser in the initial release.
- Claiming automatically extracted quantities are contractually certified without governed review.
- Storing duplicate copies of controlled model files outside Documents.
- Implementing the final universal Approvals & Workflow Engine.
- Live IoT, external digital-twin platforms or AI geometry interpretation in the initial release.
- Providing unrestricted public model sharing.

3.3 Assumptions

- Projects may contain multiple buildings, disciplines, authoring teams and coordinate systems.
- Original models may be incomplete, inconsistent or poorly classified.
- Browser delivery requires derived technical artifacts separate from controlled source documents.
- Users may need different model visibility and object data access within the same project.
- Open formats are the baseline; proprietary formats require exporters, plugins or conversion services.

4. Terminology and Core Concepts

Term	Definition
Source model	Controlled user-supplied model file or external model reference registered in Documents and BIM.
Model record	Logical BIM record containing discipline, author, coordinate, version and publication metadata.
Model version	Immutable revision of a model record with its source document revision and derived artifacts.
Federation	Published combination of multiple compatible model versions in a shared coordinate frame.
Object	Persistent element identity representing a model element, system, space or virtual spatial node.
Stable object reference	ConstructionOS identifier intended to survive model revisions when matching succeeds.
Derived artifact	System-generated geometry, tiles, metadata, indexes, thumbnails or caches used for delivery.
Viewpoint	Saved camera, visibility, clipping and selection state; may include markup and issue reference.
Digital Twin link	Typed reference from an object/spatial node to a record owned by another module.
Operational state	Visual projection of source-module status, such as progress, quality, risk or asset state.

5. Permanent Source-of-Truth Boundaries

Domain	Authoritative ownership
Documents	Original source files, controlled revisions, access, retention, legal hold and file history.
BIM / Digital Twin	Model registry, model versions, federations, conversion artifacts, object identity, normalized model properties, viewpoints, markups and visual relationships.
Projects	Project, building and project-structure records.
Units	Unit identity, commercial status, availability, price and sales data.
Work Progress	Actual completed quantities, dates, crews, accepted progress and physical execution records.
Tasks	Assigned work, due dates, accountability and execution status.
QA/QC	Inspections, NCRs, defects, quality acceptance, commissioning evidence and handover readiness.
HSE	Safety incidents, hazards, permits, restrictions and corrective actions.
Assets	Operational assets, assignment, condition, maintenance and lifecycle.
Finance / BOQ	Budgets, commitments, postings, valuations and certified financial results.
Contracts	Formal agreements, obligations and contractual records.

Domain	Authoritative ownership
Reporting	Governed metrics and issued analytical results.

File rule: Documents owns controlled source files. BIM may own only system-generated viewer artifacts and caches that are reproducible from the controlled source revision.

5.1 Write path rule

User action

- > tenant and project scope
- > authorization
- > payload and model-version validation
- > source-of-truth ownership check
- > transaction
- > audit record
- > versioned domain event
- > asynchronous processing where required

6. Personas, Roles and Access Model

6.1 Primary personas

Persona	Primary needs
Executive	Visual project health, key risks, progress, sales and high-level drill-down.
Project manager	Federated coordination, progress states, issues, decisions and package context.
BIM/VDC manager	Model registration, coordinates, federation, versions, object mapping, QA and publishing.
Designer/consultant	Discipline model review, viewpoints, BCF issues, documents and responses.
Site manager	Mobile spatial navigation, work progress, QA/QC, HSE and task context.
QA/QC inspector	Object-linked inspections, NCRs, defects and handover evidence.
HSE manager	Hazards, permits, restricted zones, emergency context and corrective actions.
Commercial/quantity surveyor	BOQ links, quantity provenance, package and cost context.
Asset/facilities manager	Asset object registry, commissioning, manuals, maintenance and telemetry readiness.
Sales/client user	Permitted unit/project visualization with restricted technical information.

6.2 Capability groups

- View published models and federations.
- View normalized properties and linked records.
- Create viewpoints and personal markups.
- Create shared markups or coordination issues.
- Register source models and versions.
- Configure coordinates and federation transforms.
- Publish or withdraw a model version/federation.

- Reconcile object identity across revisions.
- Configure status-color schemes and link types.
- Export model metadata, viewpoints or allowed geometry.
- Administer processing profiles and storage policies.

6.3 Access evaluation

```
tenant access
AND project access
AND model/federation access
AND discipline/layer/object visibility
AND field/property permission
AND linked-record permission
AND action capability
```

7. Information Architecture

7.1 Main navigation

Area	Pages
Digital Twin	Project Twin, Model Browser, Federations, Viewpoints, Issues, Progress View, Quality View, HSE View, Asset View, Unit View.
Model Management	Model Registry, Uploads/References, Versions, Processing Jobs, Validation, Coordinates, Federations, Object Reconciliation.
Administration	Format Profiles, Classification Maps, Property Rules, Color Schemes, Link Types, Storage, Performance, Permissions, Audit.

7.2 Project page integration

- Project Page includes a Digital Twin tab and summary card.
- Each building, floor, zone, unit and room may open a filtered spatial view.
- Project analytics may deep-link to a saved viewpoint or filtered object set.
- External portals use separate presentation policies and never reuse internal permissions blindly.

8. Digital Twin Spatial Hierarchy

The platform requires both imported model hierarchy and governed ConstructionOS spatial hierarchy. They may be linked but must not be assumed identical.

```
Corporate Group
-> Company
-> Project
-> Site / Parcel
-> Building / Block
-> Level / Floor
-> Zone
-> Unit / Apartment
-> Room / Space
-> System / Assembly
-> Element / Equipment
```

8.1 Spatial node requirements

- Each node has a stable ConstructionOS ID independent of any single model revision.
- Model objects may map to one or more spatial nodes.
- Virtual nodes may exist even when no model object represents them.

- Hierarchy changes are effective-dated and audited.
- Project structure remains owned by Projects/Units; BIM stores mappings and visual extents.

9. Model Registry and Model Lifecycle

9.1 Model record

Field group	Required fields
Identity	Model ID, code, name, project, discipline, model type, authoring organization.
Ownership	Responsible company, model manager, originating consultant, confidentiality.
Technical	Source format, authoring application, schema/version, units, coordinate reference.
Lifecycle	Status, current published version, superseded version, issue date, revision code.
Processing	Profile, latest job, warnings, artifact set, publish readiness.
Relationships	Building/zone scope, federation membership, linked documents, contract/package context.

9.2 Model lifecycle

```

Draft registration
-> Source attached
-> Uploaded / referenced
-> Validation queued
-> Validation passed with warnings or failed
-> Conversion queued
-> Optimized
-> Review ready
-> Published
-> Superseded
-> Archived

```

9.3 Required controls

- Published versions are immutable.
- A withdrawn version remains auditable and cannot disappear from historical links.
- Draft processing artifacts are not visible to general project users.
- Publication must not silently change the active federation without an explicit action.

10. Supported Formats and Integration Posture

Format/integration	Initial posture
IFC	Core open BIM ingestion target; support profile configured by schema and project policy.
glTF / GLB	Core optimized exchange/viewer format and supported source for prepared models.
BCF	Core issue/viewpoint exchange target.
Revit	Adapter/exporter workflow; native file processing may be deployment-specific. No direct database coupling.

Format/integration	Initial posture
Navisworks	Coordination import/export through BCF, clash data and supported exported geometry/metadata.
Unreal Engine	Future advanced ArchViz/interactive experience adapter through Datasmith/glTF/custom published package.
Cesium	Geospatial context and large-area 3D Tiles integration readiness.
ROZARIS	Future shared 3D technology and publishing adapter with strict tenant/project separation.
Point clouds	Future-ready ingestion profile for approved point-cloud formats and derived tiles.

Integration principle: Open formats are the portable contract. Proprietary integrations are adapters that may enrich the workflow but cannot become the only path to project data.

11. Model Ingestion and Validation

11.1 Ingestion methods

- Upload through Documents and register as a model source revision.
- Reference an existing controlled document revision.
- Upload through desktop authoring-tool adapter.
- Import from an approved external source in future integrations.
- Automated revision intake through a controlled service account.

11.2 Validation stages

1. File integrity, malware scan and allowed type validation.
2. Schema/parser validation.
3. Unit and coordinate extraction.
4. Hierarchy and object count analysis.
5. Global identifier and duplicate analysis.
6. Geometry validity and extreme-bound checks.
7. Property-set and classification coverage.
8. Performance estimation and processing-profile selection.
9. Project-specific mandatory-rule checks.
10. Validation report generation.

11.3 Validation results

State	Behavior
Passed	Eligible for conversion.
Passed with warnings	Conversion allowed; warnings remain visible and may block publication by policy.
Failed	No publication; issue details and remediation guidance provided.
Unsupported	Source retained in Documents but no viewer conversion attempted.

12. Coordinate Systems, Georeferencing and Units

Coordinate errors are one of the highest-risk failure modes. The module must represent source coordinates, project coordinates, shared/federation coordinates and geospatial coordinates explicitly.

12.1 Coordinate record

- Source model origin and axis orientation.
- Length and angle units.
- Project base point and survey point where available.
- Coordinate reference system identifier where configured.
- North rotation and elevation datum.
- Transform to project federation coordinates.
- Transform confidence, method and responsible user.
- Validation landmarks and tolerance.

12.2 Rules

- No hidden automatic transform may be applied without preserving the source transform.
- Federation requires compatible units and an approved transform.
- Large-coordinate models must use origin rebasing or equivalent rendering strategy.
- Geospatial conversion is versioned and reproducible.
- Measurement tools display the active unit and tolerance.

13. Model Versioning and Revision Control

13.1 Version requirements

- Immutable version ID, revision code, issue date and source document revision.
- Parent/superseded version relationship.
- Change summary and authoring organization.
- Validation report and processing profile.
- Published artifact manifest with checksums.
- Object count, added/removed/changed summary and identity-matching report.

13.2 Version states

Received -> Validated -> Converted -> Review Ready -> Published -> Superseded -> Archived

13.3 Revision comparison

- Geometry changed, property changed, classification changed, moved, added and removed states.
- Object matching confidence and unresolved mappings.
- Comparison is always between explicit immutable versions.
- Users can save comparison viewpoints and create coordination issues from differences.

14. Model Federation and Discipline Coordination

14.1 Federation record

Area	Fields
Identity	Federation ID, name, project, purpose, status and owner.
Composition	Included model versions, discipline, order, transform and visibility defaults.
Publication	Published version, publication date, audience and default viewpoint.
Performance	Scene profile, tile manifest, object count, geometry size and memory estimate.
Coordination	Clash run references, issue sets, BCF package and review status.

14.2 Federation rules

- A federation references immutable model versions; it does not copy source model data.
- Changing composition creates a new federation version.
- Discipline models may have independent publication cadences.
- Project users see only federations and components permitted to them.
- Federation transforms are explicit and auditable.

15. Conversion, Optimization and Publishing Pipeline

```
Source revision
-> secure intake
-> parse and metadata extraction
-> validation
-> object identity normalization
-> hierarchy and property index
-> geometry triangulation
-> mesh/material deduplication
-> spatial partitioning
-> LOD generation
-> compression
-> tile/package generation
-> thumbnails and manifests
-> automated QA
-> review ready
-> publish
```

15.1 Derived artifacts

- Scene manifest and dependency graph.
- Progressive geometry/3D tiles or equivalent packages.
- Metadata/property index.
- Object bounding volumes and spatial index.
- LOD and mesh compression outputs.
- Material/texture atlases where appropriate.
- Preview images and default camera.
- Checksums, processing logs and quality metrics.

15.2 Processing controls

- Asynchronous jobs, retry safety and idempotency.
- Per-tenant concurrency and resource quotas.
- Cancellation before publication.
- Worker isolation for untrusted files.
- Reproducible artifacts from source revision plus processing-profile version.
- Automated cleanup of failed temporary artifacts.

16. Stable Object Identity and Object Matching

ConstructionOS must create an internal stable object reference that is not tied only to a renderer node index or one model revision.

16.1 Identity inputs

- IFC GlobalId or equivalent native persistent identifier.
- Authoring application unique ID where available.
- Model record, object class and spatial path.
- Geometry signature and bounding box.
- Name, type, classification and selected property signatures.
- Parent/system relationships.

- Manual reconciliation decisions.

16.2 Matching outcomes

Outcome	Behavior
Exact	Stable reference automatically continues.
High confidence	Continues automatically if project policy permits; decision remains auditable.
Ambiguous	Requires BIM manager review.
Split	One prior object maps to multiple new objects; links require distribution/reconciliation.
Merge	Multiple prior objects map to one new object; links require reconciliation.
Removed	Stable reference remains historical and points to last known version.
New	New stable reference created.

16.3 Link preservation

- Operational links attach to stable object references, not transient renderer IDs.
- Every link records the model version where it was created.
- Unresolved identity changes place affected links in a reconciliation queue.
- No automated reassignment may hide low-confidence mappings.

17. Classification, Property Sets and Data Normalization

17.1 Property layers

Layer	Purpose
Raw source properties	Immutable extracted values with source names and units.
Normalized properties	Mapped ConstructionOS names, data types and units.
Derived properties	Calculated values with formula and processing version.
Operational projections	Read-only fields retrieved from connected modules.
User annotations	Authorized notes/tags that do not overwrite source model properties.

17.2 Classification mappings

- Map source classes to configured object categories, systems, trades and cost/work-package references.
- Preserve original classification values.
- Version mapping rules and show confidence/coverage.
- Allow project-specific overrides without changing global rules.
- Flag unclassified or conflicting objects.

18. Browser Viewer Architecture

18.1 Viewer modes

- Single model review.
- Federated coordination.
- Progress/status visualization.
- Quality and handover view.

- HSE/risk view.
- Asset/facilities view.
- Unit/commercial presentation view.
- Revision comparison.
- Geospatial site context.

18.2 Rendering architecture

- WebGL2 baseline with WebGPU acceleration where supported and proven.
- Progressive scene loading based on camera, visibility and priority.
- Worker-based decompression, parsing and metadata processing.
- Object-level picking through stable render mapping.
- GPU and CPU memory budgets with eviction policies.
- Device capability detection and graceful quality reduction.
- Context-loss recovery and resumable scene state.

18.3 Viewer shell

Region	Purpose
Top toolbar	Navigation modes, selection, section, measure, compare, saved view and presentation.
Left panel	Spatial tree, models, layers, filters and saved sets.
Canvas	3D scene and spatial overlays.
Right panel	Object data, links, activity, issues and source-record previews.
Bottom timeline	4D sequence, progress date and revision comparison where active.

19. Viewer Navigation and Camera Controls

- Orbit, pan, zoom, walk, fly and first-person modes.
- Fit selection, fit model, fit spatial node and home view.
- Floor/level navigation and optional minimap.
- Camera speed control and collision-aware walk mode where configured.
- Perspective and orthographic projection.
- Named standard views and view cube.
- Bookmark current camera as a viewpoint.
- Restore view after refresh or deep link.

19.1 Accessibility and usability

- Keyboard alternatives for all major viewer actions.
- Visible focus, textual object context and non-color-only states.
- Reduced-motion mode and optional camera animation disable.
- Help overlay for mouse/touch controls.

20. Object Selection, Search, Filters and Isolation

20.1 Selection

- Single, multi-select, box select and spatial-tree select.
- Parent/child selection and system selection.
- Linked-record selection from source modules.
- Selection set creation with owner and sharing scope.

20.2 Search

- Object ID, name, type, classification, property value, room, level, system and linked-record code.

- Permission-filtered results only.
- Exact, contains, range and structured property filters.
- Saved searches and selection sets.

20.3 Isolation actions

- Hide, isolate, ghost, x-ray and color override.
- Apply by object, class, discipline, system, spatial node or saved set.
- All overrides are view state only unless saved as a viewpoint/presentation state.

21. Layers, Disciplines, Worksets and Visibility

Visibility dimension	Examples
Model	Architecture, structure, MEP, facade, civil, landscape.
Discipline	Architectural, structural, mechanical, electrical, plumbing, fire, specialist.
Source layer/workset	Imported source grouping.
Classification	Walls, slabs, ducts, pipes, equipment, spaces.
Project hierarchy	Building, floor, zone, unit, room.
Operational status	Not started, in progress, completed, accepted, blocked, defective.
Confidentiality	Internal, consultant, contractor, client, restricted.

21.1 Visibility policies

- Default visibility is configurable by viewer mode and role.
- Hidden layers do not bypass object search permissions.
- Restricted layers are omitted from manifests rather than only hidden in the UI when necessary.
- Visibility state is included in viewpoints and deep links.

22. Sections, Clipping, Explode and Measurements

22.1 Section tools

- Single plane, section box and multi-plane clipping.
- Axis-aligned and user-defined clipping.
- Save clipping state in viewpoints.
- Optional cap rendering and section outline.

22.2 Explode

- Explode by hierarchy, discipline, assembly or system.
- Explode is a visual transform only and never changes coordinates or measurements.
- State may be saved in a presentation viewpoint but is excluded from issue coordinate exports unless supported.

22.3 Measurement

- Distance, angle, area, elevation and coordinate readout.
- Object-derived dimensions and geometry measurements are clearly distinguished.
- Measurements show unit, model version, tolerance and author.
- Measurements may be personal, shared or attached to an issue.

23. Viewpoints, Markups and Presentation States

23.1 Viewpoint content

- Camera and projection.
- Visible/hidden models and objects.
- Selection and color overrides.
- Section/clipping state.
- Active viewer mode and date/timeline state.
- Optional screenshot/thumbnail.
- Optional markup geometry and text.
- Linked issue, task, meeting or report.

23.2 Viewpoint scopes

Scope	Behavior
Personal	Visible only to creator.
Team/project	Visible to authorized project participants.
Issue-bound	Immutable issue snapshot plus later viewpoint revisions.
Published presentation	Controlled shared state for executive/client use.
System generated	Created by processing, comparison or reporting services.

23.3 Markups

- Pin, arrow, line, rectangle, cloud, freehand, text and measurement annotation.
- Markups use 3D anchor plus screen representation where possible.
- Deleting a shared markup follows audit and retention rules.

24. Object Data Panel and Relationship Graph

24.1 Panel sections

- Identity and current model version.
- Spatial location and hierarchy.
- Raw and normalized properties.
- Classification and systems.
- Linked tasks and work progress.
- Drawings, documents, RFIs and submittals.
- QA/QC inspections, defects, NCRs and handover records.
- HSE hazards, permits and restrictions.
- Materials, suppliers, packages and contracts.
- Asset/maintenance information where commissioned.
- Comments, meetings, decisions, risks and activity.

24.2 Relationship graph

The relationship graph is a read-only, permission-filtered view of typed links. It must show owner module, record status, link origin and effective model version. It never implies ownership transfer to BIM.

25. BCF, Coordination Issues and Clash Records

25.1 Coordination issue

Field group	Fields
Identity	Issue code, title, type, status, priority, project, federation/model version.

Field group	Fields
Spatial	Viewpoint, selected objects, location and optional clash pair.
Responsibility	Owner, assignee reference, company, discipline and due date.
Context	Description, comments, attachments, source application and BCF identifiers.
Resolution	Response, revised viewpoint, closure reason and verification.

25.2 Ownership boundary

- BIM owns coordination issue/viewpoint records and BCF exchange.
- Tasks owns executable assignments when a coordination issue requires action.
- QA/QC owns formal defects/NCRs when the issue is a quality nonconformance.
- HSE owns hazards and safety actions.
- Links preserve all relationships without duplicating the underlying record.

25.3 Clash import

- Import clash test, pair, status, distance and viewpoint from supported coordination output.
- Clash results are not automatically accepted as project issues.
- Duplicate/near-duplicate clash grouping is configurable.
- Re-running clashes preserves historical runs and issue links.

26. Tasks, Comments, Meetings and Decisions Integration

Integration	Behavior
Tasks	Create or link a task from selected objects/viewpoint; Task remains authoritative for assignment and status.
Comments	Shared contextual discussion may attach to object, viewpoint or issue using central comments infrastructure.
Meetings	Agenda/minute items may open saved viewpoints and selected object sets.
Decisions	Decision records may reference a viewpoint and object set; decision ownership remains in source module/workflow.
Mentions	User mentions publish standard notification events.

26.1 Create Task from 3D

11. Select one or more objects or a spatial node.
12. Capture current viewpoint and optional markup.
13. Enter task title, owner, due date and project context.
14. Create through Tasks API with BIM reference payload.
15. Receive Task ID and display read-only task projection.

27. Documents, Drawings and Submittals Integration

27.1 Link types

- Model source revision.
- Drawing sheet/detail.
- Specification.
- RFI and response.
- Submittal and approval status projection.

- Method statement.
- Manual, certificate and commissioning evidence.
- Photo/video and reality-capture evidence.

27.2 Rules

- Documents stores the file once; BIM stores a typed reference and spatial/object anchors.
- Opening a document rechecks current permission.
- Superseded document revisions remain traceable from historical viewpoints/issues.
- Viewer previews use safe derivatives and cannot expose restricted originals.

28. Work Progress Integration and Visual Status

28.1 Progress mapping

- Work package or activity maps to object sets, systems or spatial nodes.
- Actual quantities and dates are read from Work Progress.
- One object may have multiple progress components; visual aggregation rule must be explicit.
- Progress mapping records version, source, confidence and responsible user.

28.2 Standard visual states

State	Meaning
Not planned	No mapped schedule/work package.
Planned	Mapped but planned start is in the future.
Ready	Preconditions satisfied and work may start.
In progress	Actual execution exists but completion is below configured threshold.
Completed	Operational progress reports completion.
Accepted	QA/QC or configured acceptance source confirms acceptance.
Certified	Commercial/financial source confirms certification where applicable.
Blocked	Constraint, HSE, legal, quality or dependency blocks progress.
Unknown/stale	Source data missing or outside freshness threshold.

Progress rule: The 3D color is a projection. Work Progress owns actual completion; QA/QC owns acceptance; Finance/Contracts own certification and payment.

29. 4D Construction Sequence and Baseline Comparison

29.1 Sequence model

- Link schedule activities to objects/object sets/spatial nodes.
- Support planned start/end, actual start/end and forecast dates from the schedule/work-progress source.
- Playback by date, week, phase or milestone.
- Compare baseline, current plan and actual state.
- Filter by contractor, package, discipline, building or zone.

29.2 Timeline behavior

- Play, pause, step, date scrub and speed controls.
- Legend explains planned/actual/late/blocked states.
- Date state is included in saved viewpoints/deep links.

- Large animations update visibility incrementally to avoid scene reload.

29.3 Source boundary

BIM stores activity-to-object mapping and visual playback configuration. The scheduling and Work Progress modules own activity dates, dependencies, baselines and actual execution.

30. QA/QC, Inspections, NCRs and Handover Integration

- Pin inspections, defects, NCRs, tests, commissioning records and snags to objects or spatial nodes.
- Color objects by quality state using QA/QC-owned decisions.
- Open quality evidence and controlled documents from the object panel.
- Filter handover readiness by system, room, unit, floor or asset.
- Create inspection context from a selected object set without duplicating QA/QC records.

30.1 Quality states

Visual state	Source meaning
Inspection required	Configured QA plan requires an inspection.
Pending	Inspection/test exists but no final decision.
Passed	QA/QC decision accepted.
Failed	Inspection failed or active NCR/defect blocks acceptance.
Reinspection	Correction recorded and verification pending.
Handover ready	Required QA/commissioning evidence complete.
Handover blocked	Missing or rejected evidence remains.

31. HSE, Permits, Risks and Emergency Context

- Visualize hazards, restricted areas, permit zones, exclusion zones and emergency routes.
- Pin incidents, observations, corrective actions and toolbox-talk context where permitted.
- Show work eligibility or stop-work state without exposing restricted personal/medical details.
- Allow HSE users to create a hazard/permit context from selected spatial objects.
- Provide time-bounded overlays so expired hazards do not appear current.

31.1 HSE safeguards

- Emergency and stop-work overlays use source status and effective dates.
- No safety-critical decision may depend only on a cached model state.
- Viewer shows freshness and source timestamp for active critical overlays.
- Mobile users can access emergency context through low-detail fallback views.

32. Assets, Equipment and Operational Digital Twin

32.1 Asset-object association

- Link a commissioned asset record to one or more model objects.
- Preserve manufacturer/model/serial data in Assets; display only authorized projections.
- Link manuals, warranties, inspections, maintenance and work orders.
- Show asset condition, availability and maintenance state visually.
- Handle replacement while preserving historical object-to-asset associations.

32.2 Operational twin states

State	Example source
Installed	Work Progress / commissioning.
Commissioning pending	QA/QC commissioning record.
Operational	Assets state.
Under maintenance	Asset maintenance/work order.
Out of service	Assets/HSE restriction.
Warranty issue	Assets/Contracts/QA record.
Retired/replaced	Assets lifecycle.

33. Units, Sales and Commercial Visualization

- Link building/unit spatial nodes to Unit records.
- Visualize configured unit states such as available, reserved, sold, blocked or not released.
- Display permitted unit metadata, area, type, price and availability through Units/CRM projections.
- Use separate public/client presentation profiles with restricted technical layers.
- Support building, floor and unit selection workflows for future external portals and ROZARIS integration.

33.1 Commercial safeguards

- BIM never owns price, reservation or sale status.
- Public presentation geometry may be a separately approved model/federation.
- Internal issues, costs, defects and confidential layers must not be inferable in client mode.

34. Quantities, BOQ, Cost and 5D Readiness

34.1 Quantity types

Type	Governance
Raw model quantity	Extracted source property; preserves original name/unit.
Geometry-derived quantity	Calculated by processing service; formula, tolerance and version shown.
Mapped BOQ quantity	Association to a BOQ item or cost code.
Certified quantity	Owned by commercial/Finance/Contracts process, never inferred from geometry alone.

34.2 BOQ mapping

- Objects/object sets may map to BOQ items, cost codes, procurement packages and work packages.
- Mappings may be one-to-one, one-to-many or rule-based.
- Rule-based mapping stores the rule version and match results.
- Quantity differences between model and BOQ are reported, not silently reconciled.

34.3 Cost visualization

- Budget, committed, actual, certified and paid states remain separate.
- Cost overlays require explicit metric definitions and permission checks.
- Executive cost visualization must drill to governed Reporting/Finance sources.

35. Reality Capture, Point Clouds and Change Comparison

35.1 Reality-capture records

- Site scan/point-cloud record with date, location, coordinate reference, source document and processing status.
- Photo sphere, drone, photogrammetry or point-cloud derivative as configured.
- Permission and retention independent from the design model.

35.2 Comparison readiness

- Overlay approved reality-capture data against design/federated model.
- Visual deviation maps must show tolerance, source date and processing method.
- Automated deviation is advisory until reviewed.
- Create issues/tasks/inspections from reviewed deviations through source APIs.

36. Geospatial, Cesium and Site Context

- Position project/site models in a geographic coordinate context.
- Link parcels, roads, utilities, terrain and approved city/context layers.
- Support large-area streaming through geospatial tile adapters.
- Use origin rebasing and precision management for large coordinates.
- Separate source geospatial ownership from the BIM visualization mapping.

36.1 Cesium readiness

- Adapter supports configured 3D Tiles sources and project georeferencing.
- External tiles use signed/controlled access and attribution metadata.
- ConstructionOS object links remain on internal stable IDs even when rendered through an external geospatial engine.

37. Unreal Engine, ArchViz and ROZARIS Readiness

37.1 Advanced visualization adapter

- Publish an approved model/version/object metadata package for Unreal Engine or equivalent advanced experiences.
- Receive stable object mapping manifest back from the advanced scene.
- Separate photoreal materials/lighting from authoritative BIM geometry and properties.
- Use versioned API contracts for selections, unit status, viewpoints and deep links.

37.2 ROZARIS readiness

- Allow future publication of approved project/unit geometry and metadata to ROZARIS through an explicit adapter.
- Require independent audience, property and layer policies.
- Never expose internal ConstructionOS project records through public model IDs alone.
- Preserve provenance from public/presentation model object to internal project/unit reference where allowed.

38. Digital Twin State, Telemetry and IoT Readiness

38.1 Initial release

The initial release supports operational state projections from ConstructionOS records but does not require live sensor integrations.

38.2 Future telemetry contract

- Sensor/device registry remains owned by Assets or a future IoT domain.
- BIM stores sensor-to-object/spatial mappings and visualization configuration.

- Telemetry uses time-series references, units, quality flags and freshness thresholds.
- Critical controls must never be operated directly from the visualization without a dedicated authorized service.
- Historical telemetry overlays remain versioned and permission-aware.

39. Change Detection and Model Comparison

39.1 Comparison modes

- Model version vs previous version.
- Federation version vs previous federation.
- Discipline vs discipline where overlap analysis is useful.
- Design model vs approved reality capture.
- Baseline presentation vs current operational state.

39.2 Change categories

Category	Definition
Added	No matched prior object.
Removed	Prior object has no current match.
Moved	Transform or geometry centroid exceeds tolerance.
Geometry changed	Shape/signature changed beyond tolerance.
Property changed	Selected normalized/raw properties changed.
Classification changed	Class/system/cost/work-package mapping changed.
Unresolved	Identity matching is ambiguous.

40. Permissions, Confidential Layers and External Sharing

40.1 Security scopes

- Tenant and corporate group.
- Legal company and Joint Venture.
- Project and project role.
- Model/federation.
- Discipline/layer/workset.
- Object class and property field.
- Viewpoint/markup/issue.
- Linked operational record.
- Export or external presentation.

40.2 External sharing

- Initial release may support only authenticated ConstructionOS users.
- Future portals use purpose-built audience profiles and approved presentation federations.
- Shared links require expiry, authentication, watermarking and revocation where enabled.
- Public/unlisted links are disabled by default.

40.3 Inference prevention

- Restricted objects may be omitted from scene manifests, not merely hidden.
- Object counts, search facets, thumbnails and bounding boxes must respect restrictions.
- Linked-record previews recheck permissions at request time.
- Exports are regenerated under current permissions.

41. Notifications, Events and Scheduled Processing

41.1 Published events

- ModelRegistered, ModelVersionUploaded, ValidationCompleted, ConversionFailed, ModelReviewReady.
- ModelVersionPublished, ModelVersionSuperseded, FederationPublished, FederationChanged.
- ObjectMappingRequiresReview, ObjectLinkOrphaned, ComparisonCompleted.
- ViewpointShared, CoordinationIssueCreated, CoordinationIssueUpdated.

41.2 Notifications

The Notifications & Event Centre determines recipients, channels, grouping, escalation and delivery. BIM publishes domain events and deep-link context only.

41.3 Scheduled jobs

- Artifact integrity checks.
- Search-index refresh.
- Cache warming for active projects.
- Orphan-link detection.
- Archival and retention processing.
- Periodic source-reference validation.

42. Platform Configuration and Administration

- Enabled formats and schema profiles.
- Maximum file/object/texture sizes and processing quotas.
- Validation rules and publication blockers.
- Coordinate and unit policies.
- Classification/property normalization maps.
- Processing profiles, LOD, compression and tile settings.
- Viewer tools by role/device.
- Default federations and presentation modes.
- Color schemes and operational status mappings.
- Link types and allowed source modules.
- BCF/clash rules and issue taxonomies.
- Object-matching thresholds and reconciliation policy.
- Export permissions and watermarking.
- Retention, archive and legal-hold behavior.
- Custom fields and project-specific overrides.

Configuration rule: Every page, viewer tool, model type, link type, visual state and processing policy must honor Platform Configuration separately from user permissions.

43. Data Model

```
BimModel
BimModelVersion
BimSourceReference
BimValidationRun
BimValidationIssue
BimProcessingProfile
BimProcessingJob
BimArtifactManifest
BimArtifact
BimCoordinateSystem
```

```

BimTransform
BimFederation
BimFederationVersion
BimFederationComponent
BimSpatialNodeMapping
BimObjectStableRef
BimObjectVersion
BimObjectProperty
BimClassificationMapping
BimObjectMatchRun
BimObjectMatchCandidate
BimSelectionSet
BimViewpoint
BimMarkup
BimCoordinationIssue
BimClashRun
BimClashItem
BimObjectLink
BimProgressMapping
BimScheduleMapping
BimBoqMapping
BimAssetMapping
BimRealityCapture
BimComparisonRun
BimComparisonItem
BimViewerPreset
BimPublication
BimAuditEvent

```

43.1 Common fields

- id, tenant_id, company_id, project_id.
- status, version, created_by, created_at, updated_by, updated_at.
- confidentiality_class and access_scope.
- source_reference and correlation_id where applicable.
- record_version for optimistic concurrency.

43.2 Object link fields

Field	Purpose
stable_object_ref_id	Persistent BIM-side identity.
source_module / record_type / record_id	Authoritative linked record.
link_type	Task, document, progress, defect, asset, unit, BOQ, risk, etc.
created_on_model_version	Version where anchor was established.
effective_from / effective_to	Historical relationship validity.
link_origin	Manual, import, rule, source module or integration.
confidence / status	Confirmed, suggested, unresolved or superseded.

44. Backend Service Architecture

Service	Responsibility
BIM Registry Service	Models, versions, source references, publication and lifecycle.
Model Intake Service	Secure upload/reference validation and job creation.
Coordinate Service	Units, CRS, transforms and federation alignment.

Service	Responsibility
Model Processing Orchestrator	Pipeline state, worker scheduling, retries and manifests.
Object Identity Service	Stable references, matching and reconciliation.
Property Normalization Service	Classification/property extraction and mapping.
Federation Service	Federation composition, versions and publishing.
Scene Delivery Service	Signed manifests, tiles, metadata and cache control.
Viewpoint/Markup Service	Saved views, annotations and presentation states.
Coordination Service	BCF, issues, clash imports and viewpoint links.
Digital Twin Link Service	Typed cross-module relationships and integrity checks.
Comparison Service	Revision and reality-capture comparison jobs.
Search Service	Object/property/spatial indexing and permission filtering.
Export Service	Allowed metadata, BCF, viewpoint and geometry package exports.
BIM Audit Service	Structured read/write/publish/export audit.

45. Processing Workers and Storage Architecture

45.1 Storage classes

Storage	Content
Documents object storage	Controlled source model files and official exports.
BIM artifact storage	Derived scene packages, tiles, indexes, thumbnails and temporary processing outputs.
Relational database	Registry, versions, jobs, mappings, links, viewpoints, permissions and audit metadata.
Search index	Permission-projected object/property/spatial search documents.
Cache/CDN	Short-lived signed delivery of published artifacts.
Queue/event broker	Processing jobs and versioned domain events.

45.2 Worker classes

- Format parser/validator.
- Geometry converter and optimizer.
- Property/classification extractor.
- LOD/tiling/compression worker.
- Object matching worker.
- Comparison/clash import worker.
- Thumbnail/preview worker.
- Indexing worker.

45.3 Isolation

- Untrusted model processing runs in isolated containers/sandboxes.
- Workers use least-privilege, time/memory/disk limits and no unrestricted network access.
- Artifacts are scanned and checksum-verified before publication.

46. Domain Events and Integration Contracts

46.1 Event envelope

```
event_id
event_name
event_version
occurred_at
tenant_id
company_id
project_id
actor_user_id
aggregate_type
aggregate_id
aggregate_version
correlation_id
causation_id
payload
```

46.2 Consumed events

- ProjectStructureChanged and UnitStatusChanged.
- TaskChanged, WorkProgressRecorded and ScheduleChanged.
- DocumentRevisionPublished and DocumentAccessChanged.
- QualityDecisionChanged, DefectChanged and HandoverReadinessChanged.
- HSERestrictionChanged and PermitChanged.
- AssetStateChanged and MaintenanceChanged.
- BOQ/CostCodeChanged and ReportingMetricChanged where mapped.

46.3 Reliability

- Transactional outbox at source modules.
- At-least-once delivery with idempotent consumers.
- Unknown event versions routed safely to dead letter.
- Rebuildable projections from authoritative APIs/events.
- No direct cross-module database writes.

47. REST and Streaming API Contract

47.1 Model registry

```
GET    /bim/models
POST   /bim/models
GET    /bim/models/{id}
PATCH /bim/models/{id}
POST   /bim/models/{id}/versions
GET    /bim/model-versions/{id}
POST   /bim/model-versions/{id}/validate
POST   /bim/model-versions/{id}/process
POST   /bim/model-versions/{id}/publish
POST   /bim/model-versions/{id}/withdraw
```

47.2 Federations and scene delivery

```
GET    /bim/federations
POST   /bim/federations
POST   /bim/federations/{id}/versions
POST   /bim/federation-versions/{id}/publish
GET    /bim/scenes/{publicationId}/manifest
```

```
GET    /bim/scenes/{publicationId}/artifacts/{artifactId}
GET    /bim/scenes/{publicationId}/object-metadata
```

47.3 Objects, search and links

```
GET    /bim/objects/{stableRefId}
POST   /bim/objects/search
GET    /bim/objects/{stableRefId}/links
POST   /bim/objects/{stableRefId}/links
DELETE /bim/object-links/{id}
POST   /bim/object-matching/{runId}/resolve
```

47.4 Viewpoints and coordination

```
GET    /bim/viewpoints
POST   /bim/viewpoints
PATCH /bim/viewpoints/{id}
POST   /bim/coordination-issues
PATCH /bim/coordination-issues/{id}
POST   /bim/bcf/import
POST   /bim/bcf/export
POST   /bim/comparisons
```

47.5 API rules

- All writes require idempotency key where duplication is possible.
- All mutable records require record_version.
- Scene artifact URLs are short-lived and audience-bound.
- Large results use cursor pagination or asynchronous jobs.
- Viewer real-time job/status updates use WebSocket or Server-Sent Events.

48. Frontend Architecture and State Management

48.1 Core components

```
DigitalTwinPage
ViewerCanvas
ViewerToolbar
SpatialTree
ModelLayerPanel
ObjectSearchPanel
SelectionSetManager
ObjectDataPanel
RelationshipPanel
ViewpointManager
MarkupToolbar
IssuePanel
ProgressLegend
Timeline4D
RevisionComparePanel
SectionTool
MeasureTool
FederationManager
ModelRegistryPage
ProcessingJobPage
ObjectReconciliationPage
BimAdminPage
```

48.2 State principles

- Separate server state, scene state, viewer interaction state and persisted viewpoint state.
- URL-deep-linkable project, publication, viewpoint, selected object and viewer mode.
- Cancelable object/property queries and progressive scene loading.
- Optimistic UI only for low-risk personal state; no optimistic publish/reconcile/acknowledge actions.
- Context loss and route changes preserve recoverable view state.

48.3 Viewer state serialization

- Camera, visibility, selection, clipping, date, filters and color scheme.
- Serialized state uses a versioned schema.
- Large object sets are referenced by saved selection set rather than embedded in URLs.

49. Search Indexing and Query Architecture

49.1 Indexed fields

- Stable object ID, source object IDs, name, type and class.
- Normalized and approved raw properties.
- Project/spatial hierarchy and systems.
- Model/federation/version.
- Linked record codes and permitted status projections.
- Bounding volume and spatial search tokens.

49.2 Search rules

- Index documents are tenant/project/model permission projected.
- Confidential properties may be omitted or encrypted/separately indexed.
- Index updates are idempotent and tied to published versions.
- Unpublished drafts are visible only to authorized BIM management roles.
- Permission revocation invalidates caches and search projections.

50. Mobile, Tablet and Offline Behavior

50.1 Mobile viewer priorities

- Open project/floor/room/object from a task, inspection, defect or asset record.
- Load simplified geometry and metadata progressively.
- Select object and view linked records.
- Capture viewpoint, markup, photo and contextual task/inspection creation.
- Access HSE and emergency overlays.

50.2 Tablet

- Full viewer with adaptive side panels.
- Site inspection workflows with split 3D/data view.
- 4D and revision comparison where device capability permits.

50.3 Offline

- Offline package is an explicit, permission-scoped project subset.
- Package may include selected levels/zones, metadata, viewpoints and linked record summaries.
- Offline changes are drafts: markup, viewpoint, photo and task/inspection request payload.
- Publication, object reconciliation, federation changes and model processing require server connectivity.
- Expired/revoked packages become inaccessible after policy-defined checks.

51. Security, Privacy and Intellectual Property

- Encryption in transit and at rest.
- Signed, short-lived artifact delivery URLs.
- Least-privilege worker credentials and isolated processing.
- Malware scanning and file-type validation.
- Tenant/project/model/object/property access enforcement.
- Watermarked screenshots/exports where configured.
- Download/export disabled independently from view permission.
- No source model binaries exposed through viewer artifact URLs.
- Confidentiality labels and legal hold.
- Provider credentials stored in a secure secrets manager.

51.1 Intellectual property controls

- Model ownership, license and permitted use metadata on every source model.
- External consultants can be restricted to their own discipline or approved federation.
- Public/client presentation uses approved derivative geometry.
- Export actions record user, purpose, version and recipient scope.

52. Audit, Observability and Operations

52.1 Audit events

- Model registered, source attached, version created, validation run, processing started/completed/failed.
- Coordinate transform changed, federation changed, version published/withdrawn/superseded.
- Object mapping confirmed/rejected, object link created/removed.
- Viewpoint shared, issue created/closed, BCF imported/exported.
- Restricted model viewed, metadata exported, geometry package downloaded.

52.2 Operational metrics

- Processing queue depth and job duration by format/profile.
- Failure rate, retry rate and dead-letter count.
- Artifact size, object count, tile count and compression ratio.
- Scene first-load, first-render and interaction latency.
- GPU/CPU memory pressure and context-loss rate.
- Cache/CDN hit rate and artifact egress.
- Search latency and permission-filter failures.

52.3 Runbooks

- Failed conversion and corrupted artifact recovery.
- Coordinate/federation rollback.
- Object-link reconciliation after revision.
- Search-index rebuild.
- CDN/cache invalidation.
- Storage quota and artifact cleanup.
- Worker scaling and emergency disable of a parser profile.

53. Performance, Scalability and Availability

53.1 User-facing targets

Metric	Target behavior
Viewer shell	Interactive shell visible before detailed geometry completes.
First spatial context	Default viewpoint/bounding context appears within a few seconds on a normal broadband connection for optimized projects.
Interaction	Orbit/pan/selection remains responsive within device-quality profile.

Metric	Target behavior
Object properties	Common object data returns quickly from metadata service without loading full geometry.
Large processing	Conversion and export are asynchronous with progress and resumable/retry behavior.
Failure isolation	One failed model/component does not take down unrelated federations or viewer panels.

53.2 Scalability techniques

- Spatial partitioning and progressive streaming.
- Multiple LODs and mesh/material deduplication.
- Compressed geometry/textures and CDN delivery.
- Per-device quality profiles and memory budgets.
- Independent metadata/property service.
- Horizontal processing workers and queue partitioning.
- Published immutable artifacts for cacheability.

53.3 Availability

- Model management APIs target standard platform availability.
- Published viewer artifacts remain accessible during temporary processing-service outages.
- Stale operational overlays show freshness rather than silently disappearing.

54. Validation, Error Handling and Concurrency

54.1 User-facing error classes

Class	Behavior
Unsupported source	Keep source in Documents; explain supported conversion paths.
Validation failure	Show grouped issues, affected objects and remediation guidance.
Coordinate conflict	Block federation publication until transform is resolved.
Processing failure	Preserve logs, allow safe retry and avoid partial publication.
Permission revoked	Redact/close restricted scene or linked record immediately.
Stale record version	Return conflict and require refresh before retry.
Object mapping ambiguity	Place in reconciliation queue; do not auto-move links.
Device capacity	Switch to lower quality or simplified view with clear notice.

54.2 Concurrency

- Optimistic concurrency on model, federation, viewpoint, issue and mapping records.
- Only one active publish operation per target version.
- Processing jobs are idempotent by source checksum plus profile version.
- Concurrent viewpoint edits create versions or explicit conflicts according to sharing scope.

55. Retention, Archiving, Legal Hold and Deletion

- Published model versions and issued coordination records follow project/legal retention.
- Superseded versions may move to archive storage but remain traceable.
- Derived artifacts may be regenerated; retention policy may delete and rebuild them when source is retained.

- Legal hold blocks deletion of source references, model versions, viewpoints, issues and relevant audit records.
- Hard deletion of model registry data is restricted to exceptional administrative workflows.
- Project closure defines archive package, viewer availability and external-access shutdown.

55.1 Orphan handling

- If a linked source record is deleted according to its own policy, BIM retains a tombstone reference where legally permitted.
- If a model object is removed, historical viewpoints and links continue to resolve to the last known version.

56. Testing Strategy and Critical Scenarios

56.1 Required test layers

- Unit tests for transforms, identity matching, permission filters and state serialization.
- Parser/format contract tests with representative model fixtures.
- Pipeline integration tests for validation, conversion, artifact publication and cleanup.
- Viewer rendering, selection, clipping, measurement and context-loss tests.
- Cross-module API/event contract tests.
- Security, IP leakage, signed URL and export tests.
- Performance/load tests using large federations and concurrent viewers.
- Mobile/tablet device-matrix and offline package tests.
- Accessibility and reduced-motion tests.

56.2 Critical scenarios

1. A published architectural model is superseded while users have active tasks linked to objects.
2. An object GlobalId changes but geometry/properties strongly match the prior object.
3. One prior object is split into three objects in the new revision.
4. Two discipline models use different units and origins.
5. A user can see the federation but not the confidential structural model.
6. A user loses project access while the scene is open.
7. A conversion worker crashes after uploading some artifacts.
8. The same processing job message is delivered twice.
9. A model contains millions of objects and very large textures.
10. A project manager opens a historical viewpoint after the federation changed.
11. Work Progress says complete while QA/QC says failed.
12. A unit is sold while a client presentation is already open.
13. A scheduled activity changes after a 4D viewpoint was saved.
14. A BCF issue references an object removed in the current version.
15. A point-cloud comparison reports deviations outside tolerance.
16. A mobile device has insufficient memory for the requested model.
17. An export is requested, then the user is deactivated before completion.
18. A legal hold is applied after archival processing begins.

57. Acceptance Criteria

ID	Area	Acceptance criterion	Evidence
AC-001	Registry	Authorized users can register models and immutable versions linked to controlled Documents revisions.	Pass when verified by automated or documented QA evidence.
AC-002	Validation	Validation reports expose schema, units, coordinates, hierarchy, identity and performance findings.	Pass when verified by automated or documented QA evidence.

ID	Area	Acceptance criterion	Evidence
AC-003	Processing	A successful job creates a checksum-verified artifact manifest and no partial publication.	Pass when verified by automated or documented QA evidence.
AC-004	Publication	Only authorized users can publish or withdraw versions and federations.	Pass when verified by automated or documented QA evidence.
AC-005	Federation	A federation references explicit immutable model versions and approved transforms.	Pass when verified by automated or documented QA evidence.
AC-006	Viewer	Published federations open in-browser with progressive geometry and metadata loading.	Pass when verified by automated or documented QA evidence.
AC-007	Identity	Operational links use stable object references rather than transient renderer IDs.	Pass when verified by automated or documented QA evidence.
AC-008	Revision	Object matching reports exact, ambiguous, split, merge, removed and new outcomes.	Pass when verified by automated or documented QA evidence.
AC-009	Reconciliation	Ambiguous mappings never silently reassign operational links.	Pass when verified by automated or documented QA evidence.
AC-010	Search	Object search respects model, object, property and project permissions.	Pass when verified by automated or documented QA evidence.
AC-011	Viewer tools	Selection, isolation, sections, measurements and viewpoints function across supported device profiles.	Pass when verified by automated or documented QA evidence.
AC-012	Properties	Raw, normalized and derived properties are distinguishable and traceable.	Pass when verified by automated or documented QA evidence.
AC-013	Links	Object panel opens permission-safe projections and deep links to source records.	Pass when verified by automated or documented QA evidence.
AC-014	Progress	3D progress colors are derived from Work Progress and show freshness/source.	Pass when verified by automated or documented QA evidence.
AC-015	Quality	QA/QC states remain distinct from completion and certification.	Pass when verified by automated or documented QA evidence.
AC-016	HSE	Critical HSE overlays show source status and freshness without exposing restricted data.	Pass when verified by automated or documented QA evidence.
AC-017	4D	Users can compare baseline/current/actual sequence through linked source dates.	Pass when verified by automated or documented QA evidence.
AC-018	BCF	Viewpoints/issues can be imported/exported through the configured BCF workflow.	Pass when verified by automated or documented QA evidence.
AC-019	Documents	Original source files remain controlled by Documents; BIM stores only references and derived artifacts.	Pass when verified by automated or documented QA evidence.
AC-020	Units	Commercial unit status is read from Units/CRM and cannot be edited from BIM.	Pass when verified by automated or documented QA evidence.

ID	Area	Acceptance criterion	Evidence
AC-021	Assets	Commissioned assets can be linked to model objects without duplicating asset lifecycle data.	Pass when verified by automated or documented QA evidence.
AC-022	Security	Restricted layers/objects are not inferable through manifests, search, thumbnails or exports.	Pass when verified by automated or documented QA evidence.
AC-023	Audit	Publish, export, reconciliation and restricted-view actions are auditable.	Pass when verified by automated or documented QA evidence.
AC-024	Performance	Representative large federations meet agreed first-context and interaction targets.	Pass when verified by automated or documented QA evidence.
AC-025	Resilience	Viewer and published artifacts remain usable during isolated processing-service failures.	Pass when verified by automated or documented QA evidence.
AC-026	Offline	Approved mobile subsets can be cached and offline drafts synchronize safely.	Pass when verified by automated or documented QA evidence.
AC-027	Configuration	Project, model, viewer, processing, mapping and export behavior honors Platform Configuration.	Pass when verified by automated or documented QA evidence.
AC-028	Integration	All cross-module updates use APIs/events with no direct database coupling.	Pass when verified by automated or documented QA evidence.

58. Delivery Phases

Phase	Scope
Phase 1 - BIM foundation	Model registry, Documents links, IFC/glTF intake, validation, conversion, stable object identity, single/federated viewer, properties, search, permissions, viewpoints and audit.
Phase 2 - Connected construction twin	Tasks/Documents links, Work Progress status, QA/QC, HSE, 4D mappings, BCF issues, object reconciliation and mobile/tablet viewer.
Phase 3 - Enterprise digital twin	Assets/commissioning, Units/commercial view, BOQ/5D mappings, revision comparison, reality capture, geospatial/Cesium and advanced analytics.
Future roadmap	Unreal/advanced ArchViz, ROZARIS publication, live IoT/telemetry, automated deviation/clash assistance and external authoring-platform integrations.

58.1 Universal approval dependency

The module exposes approval hooks for model publication, external presentation, object mapping and issued coordination packages. The final cross-platform Approvals & Workflow Engine remains a later module and is not implemented here.

59. Migration and Adoption

59.1 Migration inputs

- Existing model files and revision logs.
- Project/building/floor/unit hierarchies.
- Existing BCF/clash issue packages.
- Object-to-BOQ, schedule or asset mapping spreadsheets.
- Published viewpoints and coordination records where exportable.

59.2 Migration controls

- File checksum and revision deduplication.
- Coordinate and unit validation before federation.
- Object ID coverage analysis.
- Permission and confidentiality mapping.
- Pilot federation sign-off.
- Parallel use with legacy viewer/coordination workflow before cutover.

59.3 Adoption

- Role-based viewer presets and training.
- Model publication checklist and BIM execution guidance.
- Object-linking standards and data dictionaries.
- Project-level performance budgets.
- Monthly review of orphan links, failed mappings and stale operational projections.

60. Page and Flow Registry

Page/flow	Primary purpose
Project Digital Twin	Main federated spatial experience and operational overlays.
Model Registry	Register, filter and manage project model records.
Model Version Page	Validation, processing, artifacts, changes, publish and audit.
Federation Manager	Compose versions, transforms, visibility and publication.
Processing Jobs	Track conversion, retry failures and inspect logs.
Object Reconciliation	Resolve ambiguous/split/merge mappings and affected links.
Viewpoint Library	Manage personal/shared/published viewpoints and markups.
Coordination Issues	BCF issues, clash references, status and responses.
Progress / 4D View	Timeline and status visualization from source modules.
Quality View	Inspections, defects, NCRs and handover readiness.
HSE View	Hazards, restrictions, permits and emergency context.
Asset View	Commissioned equipment and maintenance states.
Unit View	Permitted commercial unit visualization.
BIM Administration	Formats, processing profiles, mappings, colors, permissions and storage.

61. Risks and Open Decisions

Topic	Decision required
Rendering stack	Confirm WebGL2 baseline, WebGPU enhancement policy and supported browser/device matrix.

Topic	Decision required
Conversion technology	Select parser/converter stack and proprietary format deployment approach.
Initial IFC support	Define accepted schema/view profiles and project validation rules.
Artifact format	Choose scene tiling/package strategy and compression standards.
Performance budgets	Set object/triangle/texture/memory targets by device profile.
Object matching	Confirm confidence thresholds, mandatory review cases and split/merge UX.
Coordinate governance	Define project CRS/datum ownership and sign-off roles.
BCF version	Select initial import/export compatibility target and required extensions.
Offline scope	Define maximum mobile package size and expiry rules.
Client presentation	Decide whether included in BIM module or delivered only through future portals.
Reality capture	Select initial point-cloud/photo formats and processing approach.
Unreal/Cesium/ROZARIS	Define adapter milestones and licensing/deployment boundaries.

62. Definition of Done

- All must-priority functional requirements and acceptance criteria are implemented and tested.
- Source-of-truth boundaries are enforced through APIs/events and documented in code contracts.
- Model conversion and scene delivery pass representative large-project load tests.
- Stable object identity and revision reconciliation pass exact, ambiguous, split and merge scenarios.
- Permissions prevent model, object, property, linked-record and export leakage.
- Viewer controls, mobile behavior and accessibility pass the supported device/browser matrix.
- Coordinates, units, versioning, publication and rollback procedures are production ready.
- Processing, storage, search, CDN, monitoring, retry and disaster-recovery runbooks are approved.
- No unresolved critical security, intellectual-property, data-loss or coordinate-integrity defects remain.

Final product position: The BIM, 3D & Digital Twin Module is the governed spatial operating layer that connects every physical model object to the authorized ConstructionOS records that describe its design, execution, quality, safety, commercial and operational state.